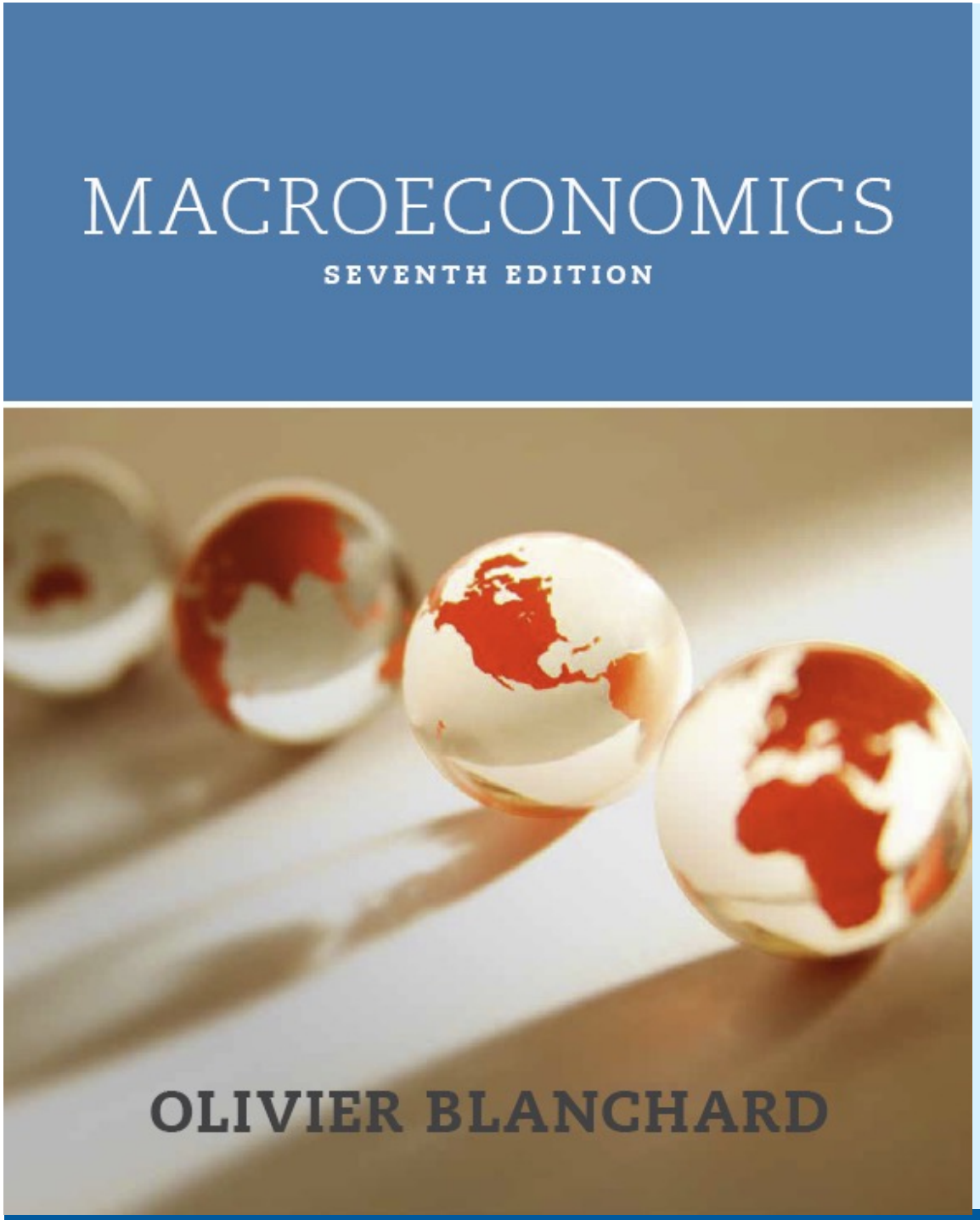


The Goods Market

Chapter 3

ALWAYS LEARNING
PEARSON

Facts about the Business Cycle

Short-run fluctuations in the growth rate of output and employment are called the **Business Cycle.**

1. Economic **booms/Expansions** are periods of expansion of GDP (positive growth), associated with increasing employment and declining unemployment.

2. **Recessions** are periods (lasting at least two quarters) in which real GDP falls (negative growth).

mruniversity introduction-business-cycle-fluctuations [3 min]

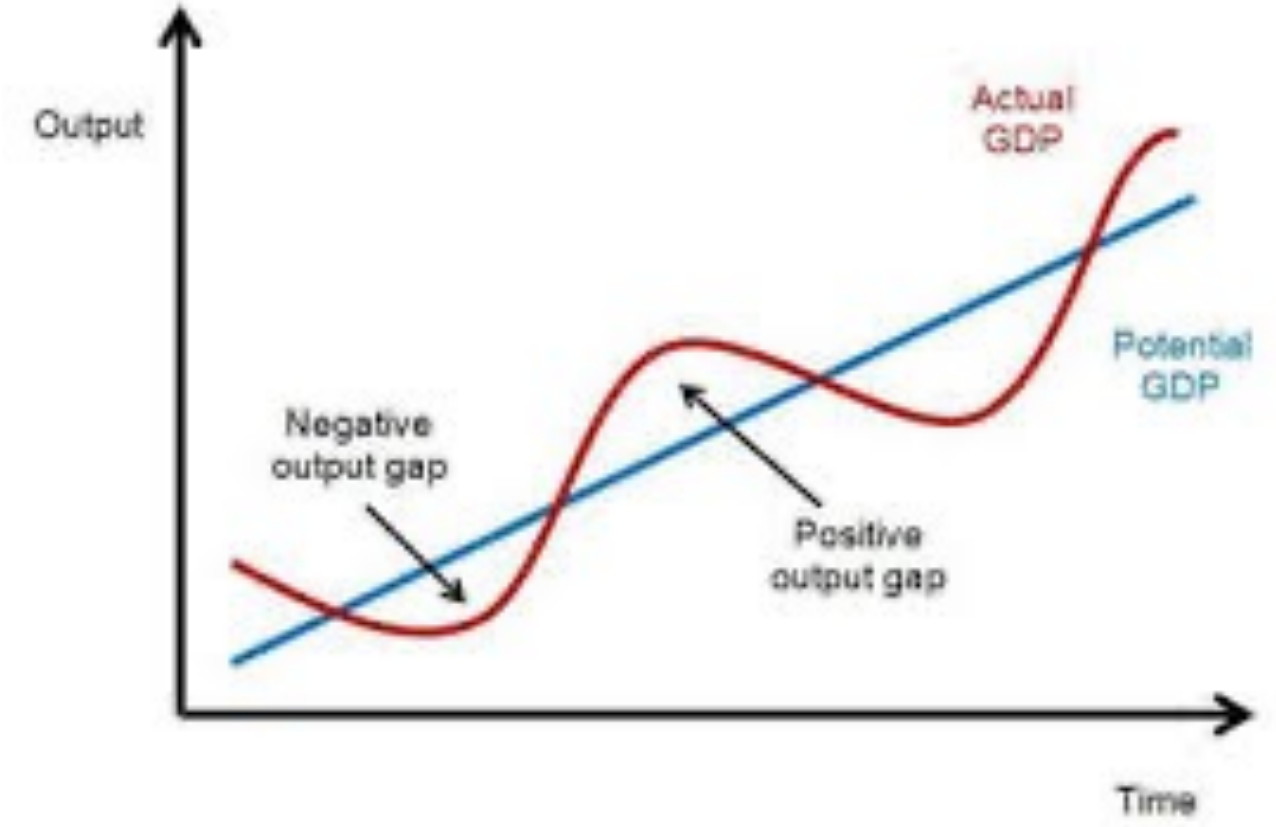
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Output gap

Country	Potential GDP growth (2026)
Spain	2.5%
United States	≈2.5%
China	≈4.3%

Source: IMF

It is an economic measure of the difference between the **actual** output for an economy and its **potential** output.



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Chapter 3 Outline

The Goods Market

3-1 The Composition of GDP

3-2 The Demand for Goods

3-3 The Determination of Equilibrium Output

3-4 Investment Equals Saving: An Alternative Way of Thinking about Goods—Market Equilibrium

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The Goods Market

- When economists think about year-to-year movements in economic activity, they focus on the interactions among *production*, *income*, and *demand*:
  - Changes in the demand for goods lead to changes in production
  - Changes in production lead to changes in income
  - Changes in income lead to changes in the demand for goods

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The composition of GDP: US, 2025

	Component	Billions of dollars	Percent of GDP
	GDP (Y)	30,762.1	100.0
1	Consumption (C)	20,954.9	**68.1**
2	Investment (I)	5,443.5	17.7
	Nonresidential	4,250.8	13.8
	Residential	1,192.7	3.9
3	Government spending (G)	5,275.1	**17.1**
4	Net exports (NX)	−926.5	−3.0
	Exports (X)	3,319.7	10.8
	Imports (IM)	−4,246.1	−13.8
5	Inventory investment	15.1	0.05

Source: Federal Reserve

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The composition of GDP: China, 2025

	Component	Billions of dollars	Percent of GDP
	GDP (Y)	≈18,750	100.0
1	Household consumption (C)	≈7,487	**39.9**
2	Gross capital formation (I)	≈7,594	**40.5**
3	Government consumption (G)	≈3,118	16.6
4	Net exports (NX)	≈533	2.8
	Exports (X)	≈3,754	20.0
	Imports (IM)	≈−3,221	−17.2

Source: World Bank

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Class Activity

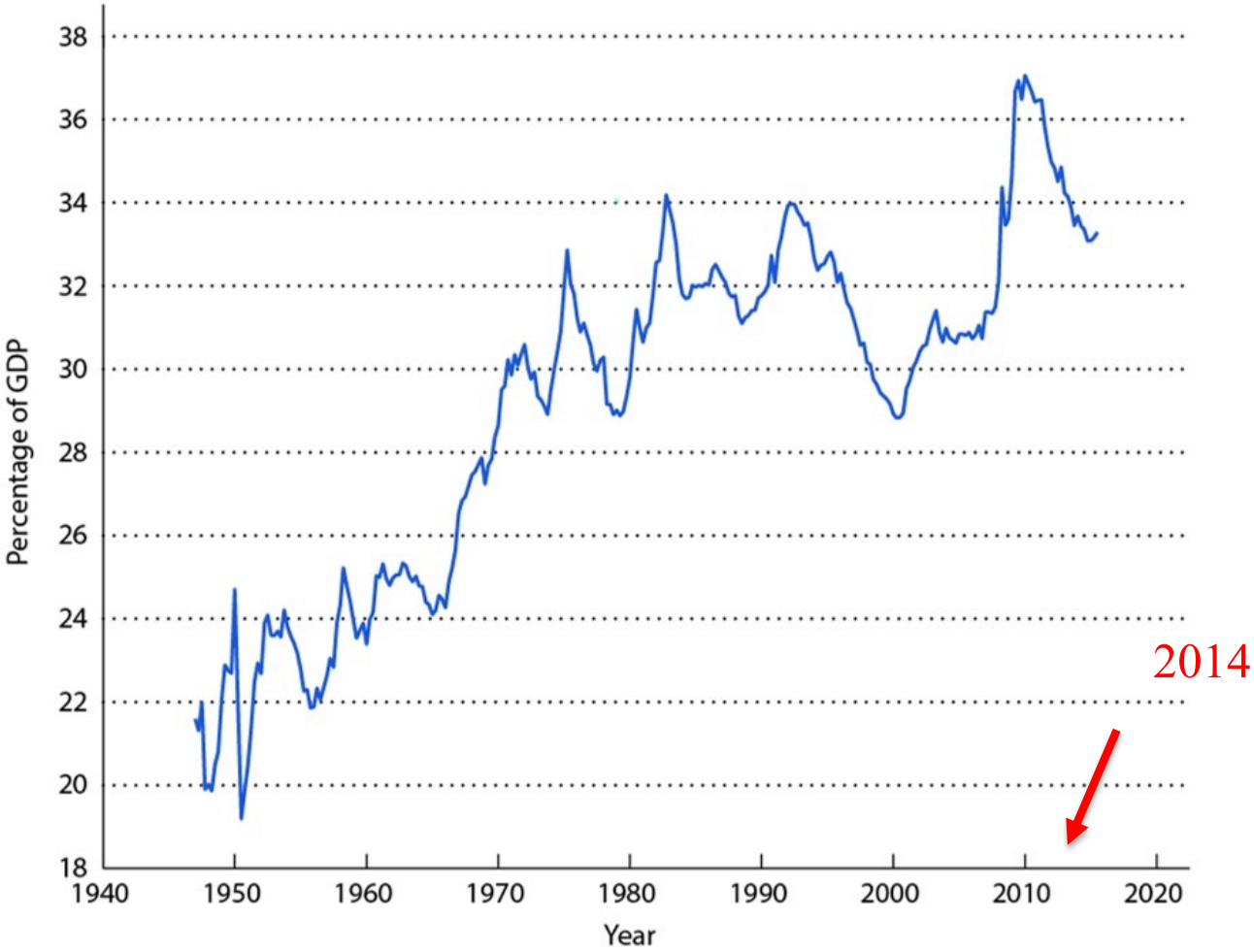
- Go to: https://stats.oecd.org/Index.aspx?DatasetCode=SNA_TABLE1
- Download the Excel file with the correct data to compare GDP internationally.
- Which is the description of the data that you chose?

GDP, REAL, US$ OR INTERNATIONAL DOLLARS

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3-1 The Composition of GDP

U.S. Total Government Outlays as a Percentage of GDP



It is not the same as Government Spending (G)!!

Source: Williamson (2017)

https://data.oecd.org/gga/general-government-revenue.htm

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In what should Governments spend and why?



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Class Activity

- Go to: https://data.oecd.org/gga/general-government-spending.htm#indicator-chart
- The graph shows Government spending per function
- In which functions do Governments spend more?

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3-2 The Demand for Goods

$$Z \equiv C + I + G + X - IM$$

- The above identity defines the total demand for goods (Z) as consumption, plus investment, plus government, plus export, minus imports.

- In a closed economy ($X = IM = 0$):

$$Z \equiv C + I + G$$

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3-2 The Demand for Goods

- Consumption (C) is a function of disposable income (Y_D), which is the income that remains once consumers have received government transfers and paid their taxes.

$$C = C(Y_D) \qquad (3.1)$$
$$(+)$$

- $C(Y_D)$ is called the **consumption function**.
- This is a **behavioral equation** that captures the behavior of consumers.

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3-2 The Demand for Goods

- Assume that the consumption function is a **linear relation** with two **parameters**, c_0 and c_1:

$$C = c_0 + c_1 Y_D \tag{3.2}$$

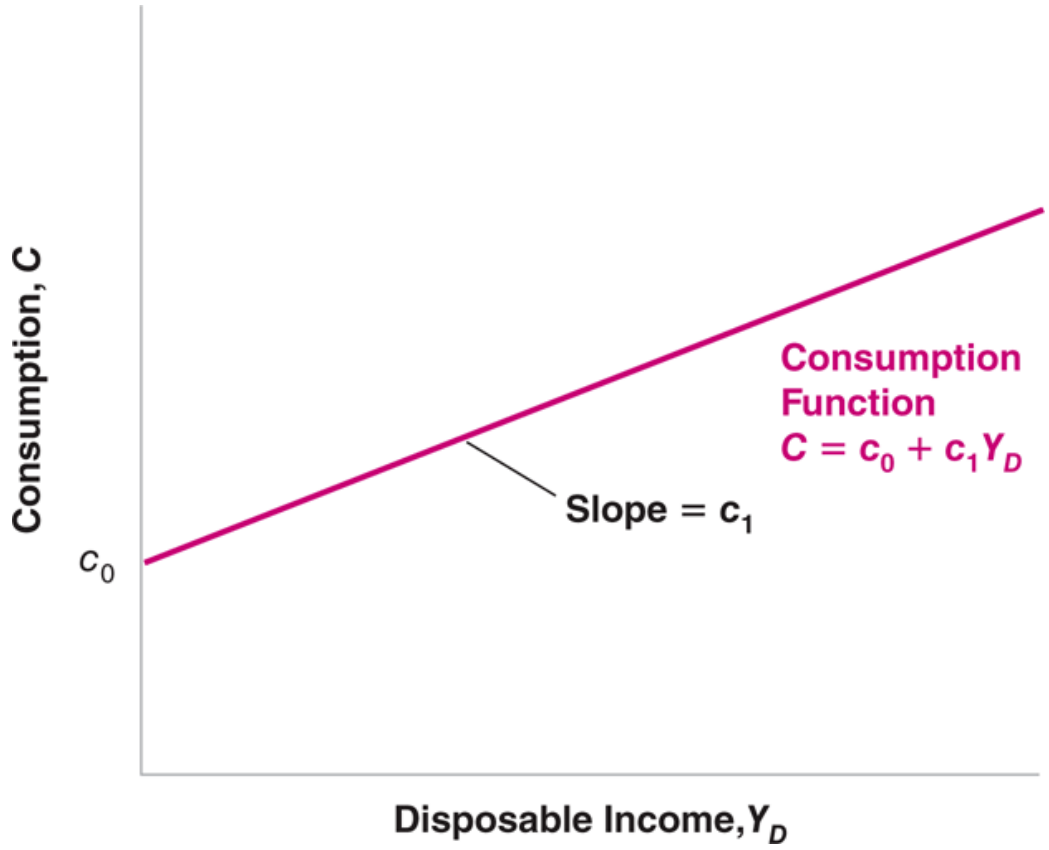
- c_1 is the **propensity to consume**.
- c_0 is what people would consume if their disposable income equals zero.
- Changes in c_0 reflect changes in consumption for a given level of disposable income.

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3-2 The Demand for Goods

Figure 3-1 Consumption and Disposable Income

Consumption increases with disposable income but less than one for one.
A lower value of c_0 will shift the entire line down.



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3-2 The Demand for Goods

- Disposable income is:

$$Y_D \equiv Y - T$$

where Y is income and T is taxes minus government transfers.

- Replacing Y_D in equation (3.2) gives:

$$C = c_0 + c_1(Y - T) \tag{3.3}$$

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3-2 The Demand for Goods

- **Endogenous variables**: variables depend on other variables in the model

- **Exogenous variables**: variables not explained within the model but are instead taken as given

$$I = \bar{I} \tag{3.4}$$

- A bar on investment means investment is taken as given.

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3-2 The Demand for Goods

- *T* and *G* describe **fiscal policy**—the choice of taxes and spending by the government.

- *G* and *T* are assumed exogenous because:
  - Governments do not behave with the same regularity as consumer or firms.
  - This book will typically treat *G* and *T* as variables chosen by the government and will not try to explain them within the model.

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3-3 The Determination of Equilibrium Output

- Assume $X=IM=0$, so

$$Z \equiv C + I + G$$

- Replacing C and I from equations (3.3) and (3.4):

$$Z = c_0 + c_1(Y - T) + \bar{I} + G \tag{3.5}$$

- **Equilibrium in the goods markets** requires

$$Y = Z \tag{3.6}$$

- This is an **equilibrium condition**.

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3-3 The Determination of Equilibrium Output

- Replacing Z in (3.6) by equation (3.5) gives

$$Y = c_0 + c_1(Y - T) + \bar{I} + G \qquad (3.7)$$

- *In equilibrium, production (Y) is equal to demand, which in turn depends on income (Y), which is itself equal to production.*

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3-3 The Determination of Equilibrium Output

- Macroeconomists always use three tools:

  1. Algebra to make sure that the logic is correct
  2. Graphs to build the tuition
  3. Words to explain the results

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3-3 The Determination of Equilibrium Output

- Rewrite equation (3.7):

$$Y = c_0 + c_1 Y - c_1 T + \bar{I} + G$$

- Reorganize the equation:

$$(1 - c_1)Y = c_0 + \bar{I} + G - c_1 T$$

- Divide both sides by $(1 - c_1)$:

$$Y = \frac{1}{1 - c_1}[c_0 + \bar{I} + G - c_1 T] \tag{3.8}$$

which characterizes equilibrium output in algebra.

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3-3 The Determination of Equilibrium Output

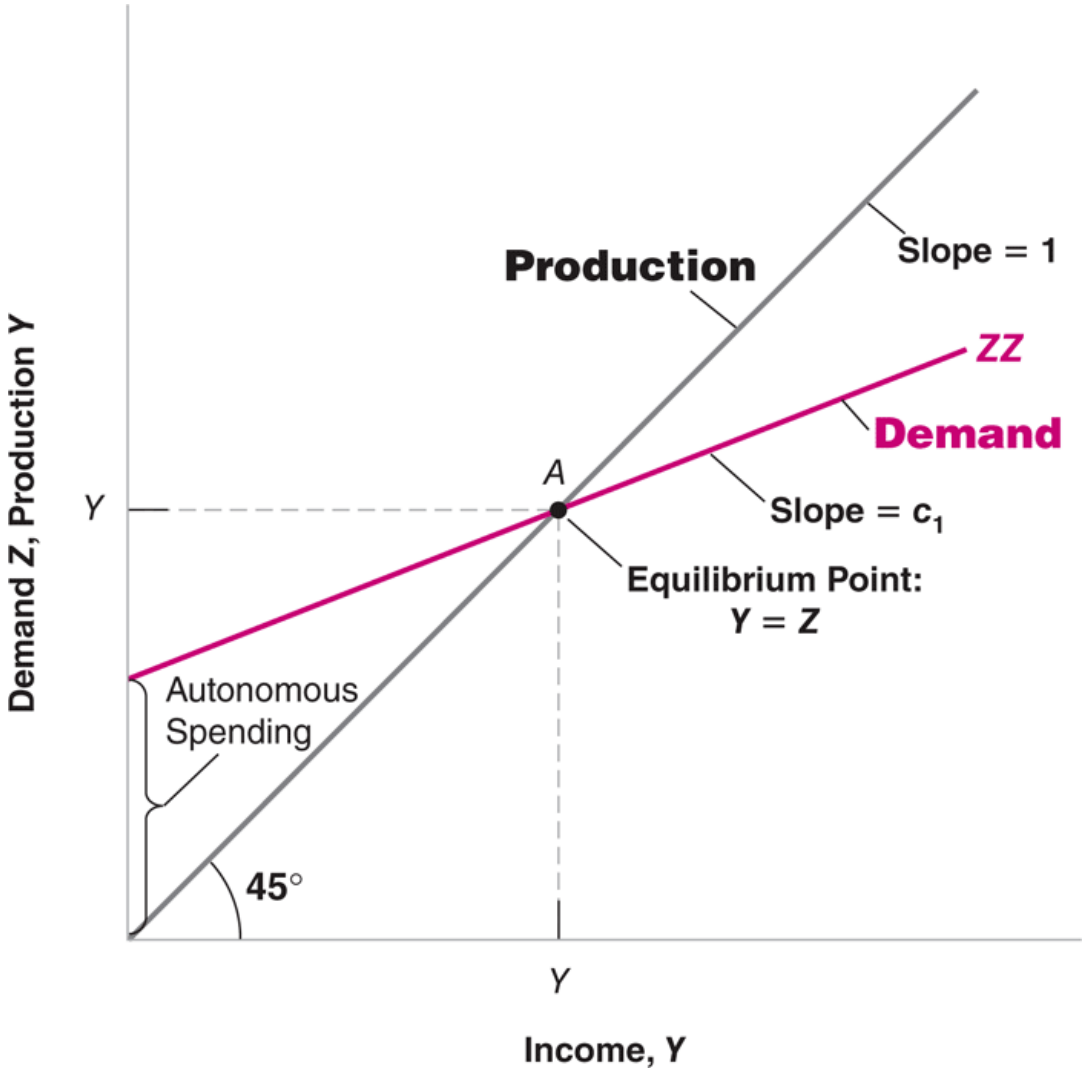
- **Autonomous spending**: $[c_0 + \overline{I} + G - c_1 T]$
- Autonomous spending is positive because if $T = G$ (**balanced budget**) and $c_1$is between 0 and 1, then $(G - c_1 T)$ is positive, and so is autonomous spending.
- The term 1/(1-c_1) is the **multiplier**, which is larger when c_1 is closer to 1.
- If c_1 equals 0.6, the multiplier equals 1/(1 – 0.6) = 2.5, meaning that an increase of consumption by $1 billion will increase output by 2.5 x $1 billion = $2.5 billion.

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3-3 The Determination of Equilibrium Output

Figure 3-2
Equilibrium in the Goods Market

Equilibrium output is determined by the condition that production is equal to demand.



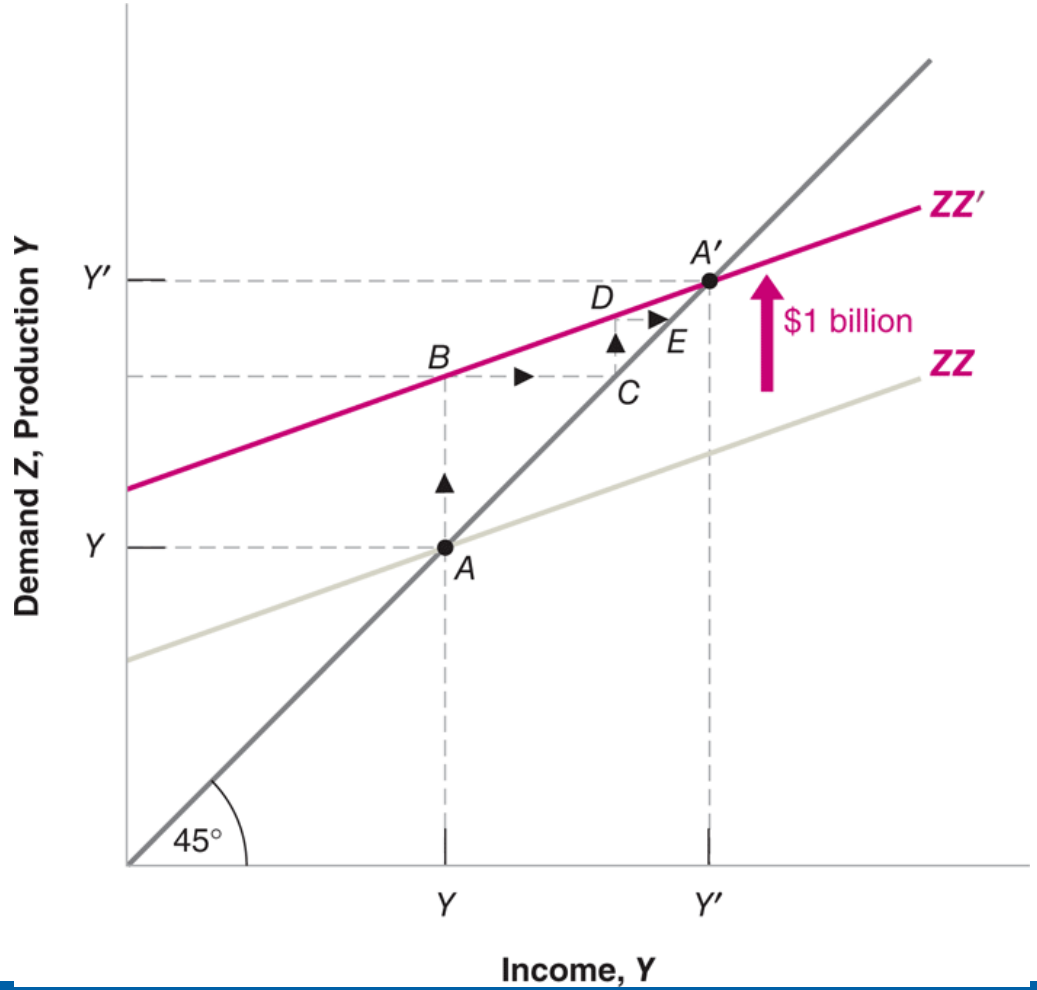
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3-3 The Determination of Equilibrium Output

- Suppose c_0 increases by \$1 billion.

Figure 3-3 The effects of an increase in autonomous spending on output

An increase in autonomous spending has a more than one-for-one effect on equilibrium output.



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3-3 The Determination of Equilibrium Output

- *AB*: first-round increase in production
- *BC*: first-round increase in income
- *CD*: second-round increase in demand
- *DE*: second-round increase in production and income
- The total increase in production after n+1 rounds:

$$1 + c_1 + c_1^2 + \ldots + c_1^n$$

which is a **geometric series** with a limit of $1/(1-c_1)$.

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3-3 The Determination of Equilibrium Output

- The adjustment of output over time is called the **dynamics** of adjustment.

- How long the adjustment takes depends on how and when firms revise their production schedule.

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Debate Question:

- In the aftermath of the 2008 financial crisis, countries like Greece faced collapsing output, high unemployment, and ballooning debt.
- European institutions pushed for **austerity measures**—cutting government spending and raising taxes—to restore fiscal balance.
- Critics, however, argued that what Greece needed was the opposite: more government spending to boost demand, even if that meant higher deficits in the short term.
- If you were advising the Greek government in 2010, would you support austerity to stabilize public finances, or a Keynesian stimulus to revive growth and employment?

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Debate Question:

- *After the COVID-19 pandemic, the US massively increased government spending through stimulus checks, expanded unemployment benefits, and public investment programs.*
- *Supporters argued this Keynesian response was essential to avoid a depression and speed up recovery.*
- *Critics countered that such high spending fueled inflation and worsened public debt.*
- *If you were advising the U.S. government in 2021–2022, would you have supported large-scale fiscal stimulus despite rising debt, or favored fiscal restraint to prevent inflation and stabilize finances?"*
-

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3-4 Investment Equals Saving: An Alternative Way of Thinking about Goods—Market Equilibrium

- John Maynard Keynes articulated an alternative model that focuses instead on investment and saving in the General Theory of Employment, Interest and Money in 1936.
- **Private saving** (S) is

$$S \equiv Y_D - C$$

$$S \equiv Y - T - C$$

- By definition, **public saving** = $T - G$.
- Public saving > 0 **Budget surplus**
- Public saving < 0 **Budget deficit**

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3-4 Investment Equals Saving: An Alternative Way of Thinking about Goods—Market Equilibrium

- In equilibrium:

$$Y = C + I + G$$

- Subtract T from both sides and move C to the left side:

$$Y - T - C = I + G - T$$

- The left side of the equation is simply S, so

$$S = I + G - T$$

- Or equivalently

$$I = S + (T - G) \qquad (3.10)$$

- This is the ***IS relation***, which stands for "**I**nvestment equals **S**aving".

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3-4 Investment Equals Saving: An Alternative Way of Thinking about Goods—Market Equilibrium

- Two equivalent ways of stating the condition for equilibrium in the goods market:

Production = Demand

Investment = Saving

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3-4 Investment Equals Saving: An Alternative Way of Thinking about Goods—Market Equilibrium

- We can also derive equation (3.8) using equation (3.10).
- Because consumption behavior implies that:

$$S = Y - T - C$$

$$= Y - T - c_0 - c_1(Y - T)$$

Rearranging terms, so

$$S = -c_0 + (1 - c_1)(Y - T) \qquad (3.11)$$

- $(1-c_1)$ is called the **propensity to save**, which is between zero and one.

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3-4 Investment Equals Saving: An Alternative Way of Thinking about Goods—Market Equilibrium

- In equilibrium, $I = S$, so that equation (3.10) becomes:

$$I = -c_0 + (1 - c_1)(Y - T) + (T - G)$$

- Solve for output:

$$Y = \frac{1}{1 - c_1}[c_0 + \bar{I} + G - c_1 T] \qquad (3.12)$$

which is the same as equation (3.8).

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